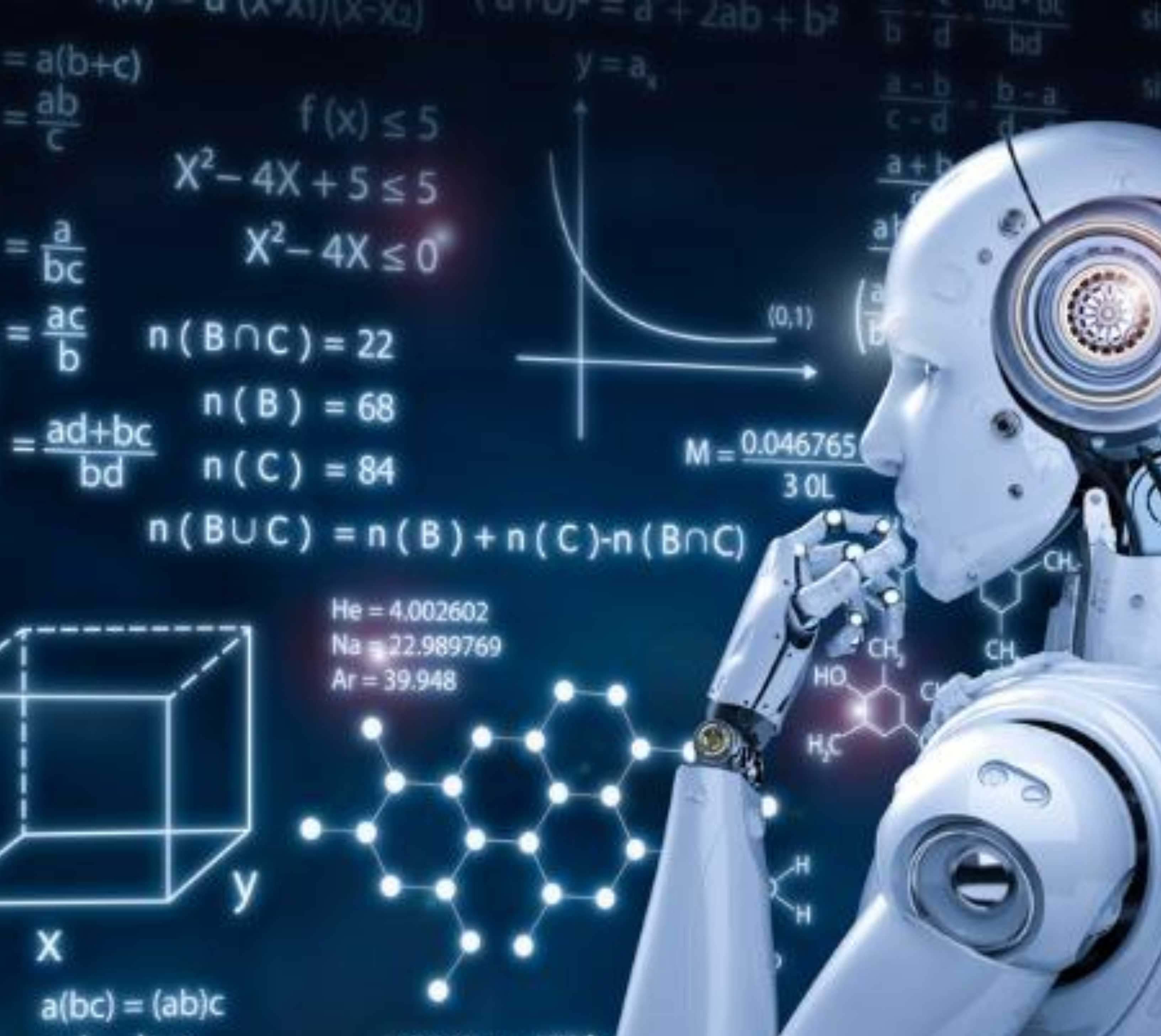
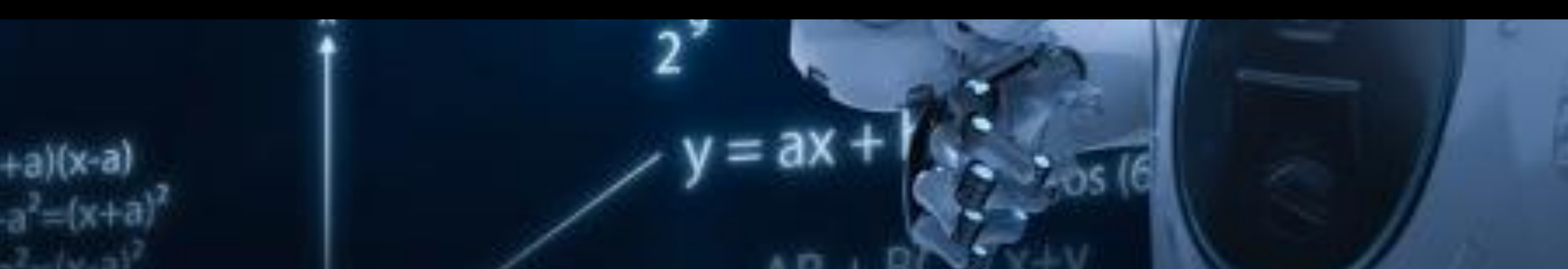


Machine Learning Task – 2

IEEE SSCS – Level 1



Task 1 :

K-Nearest Neighbors (KNN) Classification Task

In this task, you will implement a K-Nearest Neighbors (KNN) classifier to solve a supervised classification problem using the dataset provided in the link below.

The objective is to investigate how the choice of the hyperparameter K affects the classification performance.

You should begin by loading and preprocessing the dataset, handling any missing values if necessary, encoding categorical variables, and splitting the data into training and testing sets. Appropriate feature scaling (such as StandardScaler) must be applied before training the KNN model, since KNN relies on distance calculations.

Train the classifier for every value of K ranging from 2 to 150, and evaluate the model using the F1-score on both the training and testing datasets.

Store the F1-score corresponding to each K value, then create a single graph with K on the x-axis and

F1-score on the y-axis, plotting separate curves for the training and testing scores. Analyze the resulting curves to identify signs of overfitting (high training performance but low testing performance) and underfitting (low performance on both datasets), and determine the value of K that provides the best generalization performance. Finally, summarize the observations, discuss the trade-off between model complexity and generalization, and justify the selection of the optimal K based on the experimental results.

Hint : use sklearn in this task, don't implement it from scratch

Dataset:
https://www.kaggle.com/datasets/uciml/mushroom-classification

Task 2 :

Logistic Regression Task : Learning Rate Analysis

In this task, you will implement a Logistic Regression classifier using the dataset provided in the link Begin by loading and preprocessing the dataset, handling missing values if necessary, encoding categorical features, splitting the data into training and testing sets, and applying feature scaling using StandardScaler. **Train the Logistic Regression model using different learning rate values ranging from 0.001 to 1.000 with a step size of 0.001, while keeping the remaining hyperparameters fixed. For each learning rate, evaluate the model using the F1-score on both the training and testing datasets.** Store the results and create a graph with Learning Rate on the x-axis and F1-score on the y-axis, plotting separate curves for the training and testing scores. Analyze how the learning rate influences the model's convergence and classification performance, then identify the learning rate that achieves the best balance between training and testing performance.

Task 3 :

Logistic Regression Task: Number of Iterations Analysis

Using the same preprocessed dataset, investigate the effect of the **maximum number of iterations** on the performance of the Logistic Regression model. Train the classifier using different iteration values ranging from **1,000 to 100,000**, while keeping all other hyperparameters fixed. For each iteration value, compute the **F1-score** on both the training and testing datasets. Store the evaluation results and create a graph with **Number of Iterations** on the x-axis and **F1-score** on the y-axis, plotting separate curves for the training and testing scores. Analyze the resulting curves to determine whether increasing the number of iterations improves convergence and model performance or reaches a point where additional iterations provide little or no benefit. Finally, identify the iteration value that offers the best trade-off between computational cost and classification performance.